

Pergunta 1

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

When compared to other engineering disciplines, with many, many years, software engineering is really young. Although the term 'software engineering' is commonly known today, the very first time it was used with relevance was less than one century.

When and why?

- a) in the late 30's, to improve the development of computer software
- b) in the late 60's, to address the growing importance of software
- c) in the 90's, to compile the history of software and its practices, languages, and tools
- d) in the 90's, to make a distinction between the science and the engineering of software

Option b)

The term software engineering only started to be used after two conferences sponsored by NATO as a response to the software crisis since the development of software didn't have any specified good practices which led to a lot of failures and money lost in the process. This two conferences intended to set some ground rules for software engineering with the purpose to make it easier to develop software.

Pergunta 2

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

RUP is composed by several engineering disciplines. Analysis and design is one of those disciplines. What roles are required by this discipline? Elaborate what they use as input and what is the output of their work.

- a) The whole development team;
- b) Lead developer and clients;
- c) Requirements engineers and product manager;
- d) Software architects and designers.

Option d)

In the analysis and design discipline, the architects start by performing an architectural analysis giving it to the designers for them to make a use-case analysis. After this, the architects make develop the architectural design and describe both the concurrency an the distribution, finalising with a review of their work. This review is used by the designers to make the subsystem and the class design, which will lead them to the use-case diagram finishing with another review.

Pergunta 3

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

Regarding use case modelling, an *extend* relationship aims to (justify your answer):

- a) Distinguish what is mandatory from what is optional, regarding use cases
- b) Show a mandatory relationship that always happens, regarding use cases
- c) Show a mandatory relationship that always happens, regarding actors
- d) Be represented as a straight solid line.

Option a)

The extend relationship allows the highlight of optional features and distinguishes what is mandatory or essential from what is optional. This extensions indicate some conditional behavior from the base case, which should make sense from itself.

Pergunta 4

Respondida

Pontuou 2,00 de 2,00

Destacar pergunta

The Unified Modeling Language was created to be a standard language that engineers could use to specify and discuss software decisions. Several diagram types are available for capturing different information.

Considering the need to describe a program's structure, which diagrams should be used?

- a) Class diagram and sequence diagram;
- b) Component diagram and class diagram;
- c) Sequence diagram and use-case diagram;
- d) Component diagram and sequence diagram.

Option b)

The component diagram is used to show how the software is decomposed for development, into software components. The class diagram shows the classes and their connections in the program. Using both of these types of diagrams is the best and easiest way to describe the program's structure.

Pergunta 5

Respondida Pontuou 2,00 de 2,00 Destacar pergunta

The Agile manifesto defends:

Individuals and interactions over processes and tools
Working software over comprehensive documentation
Customer collaboration over contract negotiation
Responding to change over following a plan

In your opinion, what did the authors wanted to achieve by introducing the manifesto? Justify your option.

- a) A faster way to develop software, by discarding the need for documenting requirements and source code;
- b) A more efficient software development, by adopting novel development architectures and tools;
- c) A cheaper way to develop software, by eliminating overheads and unnecessary roles;
- d) A less bureaucratic approach to develop software by bringing clients and development team to work better and together.

Option d)

The agile manifesto was focused on bringing clients and development team together, as the first line says: "Individuals and interactions over processes and tools", to work on the software in a way that if any requirement wasn't met, the client could quickly give feedback to the developers so that they could fix it. This continuous feedback from the client on what is being done not also makes it easier to fix any problems from the start, but also makes the development team work harder and better in order to satisfy the client and receive only positive feedbacks, ending up with a better build and developed software.

Pergunta 6

Respondida Pontuou 2,00 de 2,00 Destacar pergunta

eXtreme Programming is an iterative methodology that values simplicity and adaptability to change. Despite its advantages, it is not ideal for all software projects. Choose which of the following projects you believe XP would not be applicable to, justifying your choice.

- a) Designing a large corporation website;
- b) A project with a strict deadline and a specific set of features;
- c) A research project for a heart implant;
- d) A large scale micro-service cloud application.

Option c)

eXtreme Programming is not very appropriate for safety critical systems where every change made to the software has to be done very carefully to preserve safety. In a heart implant project, one of the key aspects of the software must be is safety, or, in other words, the absence of errors and bugs. Any bug in a software of this kind could have tragic results. This is where XP is not as appropriate to this kind of projects, as the tests made only show the presence of errors, not the absence of them. To ensure safety, the software should be meticulously done and developed.

Pergunta 7

Respondida Pontuou 2,00 de 2,00 Destacar pergunta

Reverse engineering (please justify your answer):

- a) Is a reengineering process activity
- b) Means converting code to a new language
- c) Is to clean-up and restructure system data
- d) Is to reorganise the program structure.

Option a)

Reverse engineering is a reengineering process where the team analysis the current program in order to understand it, how it works, and how everything is done. This process is key for the reengineering process because in order to make a program better, we first need to understand how it was done, so we can come up with better solutions.

Pergunta 8

Respondida Pontuou 2,00 de 2,00 Destacar pergunta

The Project Management Body of Knowledge states that the project management processes are divided into process groups.

How many process groups are there and what are their names and main objective:

- a) 6
- b) 12
- c) 5
- d) 3

Option c)

There are five process groups in the project management processes. The Initiating Process Group, the Planning Process Group, the Executing Process Group, the Monitoring and Control Process Group and finally the Closing Process Group. The Initiating, as the name says, is mostly active in the beginning of the project and the Closing at the end, whereas the Planning, the Executing and the Monitoring and Control Process Groups are active throughout the whole project. The Planning process group is mostly responsible for planning the project, the Executing is responsible for the project execution and development and the control process group is responsible for ensuring that everything is working as expected.

Pergunta 9

Respondida

Pontuou 1,80 de 2,00

🚩 Destacar pergunta

Briefly describe the most important phases and respective activities of RUP that you have used in your app project, explaining their purpose in the project and the corresponding results.

In the development of our app, the most important phases of RUP used were the inception and the construction phases. The first one, the iteration 0, was where we established the project's main purpose and what we were going to do or not in future iterations, or, in other words, we started our Business Modelling. We also set the use cases for the software and their acceptance tests, making an estimate of the cost it would take us to develop those features and checked if our given iterations were a reasonable amount for those user stories, the Requirements, and started to think of possible architectures that could be developed in our software. The other important phase was the construction phase, the other iterations, where we developed and tested our features. In this phases, the activities were Implementation and Test, in the majority, and some Analysis and Design and Requirements as we needed to adapt some previously defined aspects of our project in order to meet with the resources we had.

Pergunta 10

Respondida

Pontuou 1,50 de 2,00

🚩 Destacar pergunta

Briefly describe the most important agile practices that you have used in your app project, explaining their purpose in the project and the corresponding results.

The development of this project by itself was using one of the main agile practices which was developing the software incrementally, which was done in iterations. The iterations were short (2 weeks each) which resulted in quick feedback from the client (in this case, the teacher) on what was being done which is another one of the main practices of agile. This quick feedback made it easier to fix any bugs from the start, and made it possible to redesign our code and adapt it to changes in the requirements. The interaction with the client was crucial for this project since we were motivated to be more hardworking and produce better software just to satisfy the client every iteration.

The software changed a lot throughout the whole project, but as we used agile practices, such as the interaction with the teacher every week and the incremental design, it turned out a very well built software that met all the client's requirements.